LEONARDO COLÒ

NUMBER THEORY & CRYPTOGRAPHY

DOCTOR OF PHILOSOPHY IN MATHEMATICS

Date of Birth: 14 oct. 1994
Nationality: Italian
Address: 200 University Ave. W.,
Office MC 5438,
Waterloo, ON, N2L 3G1, Canada
Email: lcolo@uwaterloo.ca
Webpage: leonardocolo.com

SUMMARY

I am currently a Postdoc in the Department of Combinatorics and Optimization at the University of Waterloo under the mentorship of Prof. David Jao. My broad research interests lie at the crossroad of number theory and cryptography. In particular, I investigate various aspects of supersingular isogeny graphs and exploit their potential new cryptographic applications.

WORK EXPERIENCE

Summer 2026 **Internship.** Qorsa, Waterloo, ON Canada.
- Building a Go-based TLS scanner for post-quantum migration.

2023-2026 **Postdoctoral Fellow.** University of Waterloo, ON Canada.
- Program: *Accelerating the transition to quantum-resistant cryptography*.

2021-2023 **ATER.** Aix-Marseille Université.
- Fixed-term, full-time teaching and research position.

EDUCATION

2018-2022 **PhD in Mathematics and Cryptography.** Aix-Marseille Université.
- Advisor: *Prof. David Kohel.*
- Thesis Title: *Oriented supersingular elliptic curves and class group actions.*

2016-2018 **Master Degree in Algebra and Number Theory.** ALGANT Master Program.
- First year at Concordia University, Montréal (QC, Canada). **Cumulative GPA: 4.14/4.00**
- Second year at Università degli Studi di Milano, Milano (Italy). **Grade: 110/110 cum Laude**
- Thesis: *p-adic Abelian Integrals: from Theory to Practice,*

2013-2016 **Bachelor Degree in Mathematics.** Università degli studi di Milano.
- Final Seminar: *Intersection of Ideals in a Multivariable Polynomial Ring.* **Grade: 110/110**

2008-2013 **High School Diploma.** Liceo Scientifico Statale “Leonardo da Vinci”, Milano (Italy).
- Final research: *Mathematical models for the theory of conflicts*. **Grade: 100/100**

AWARDS

Honors Member of the Concordia University Chapter of the Golden Key International Honour Society.

Scholarships
- ”Progetto Eccellenze” awarded by Città di Pioltello, a.y. 2012/2013 & 2016/2017.
- ”Concordia International Tuition Award of Excellence” awarded by Concordia University, a.y. 2016/2017.

PUBLICATIONS

2026 **Zero-Knowledge proofs of isogeny diamonds.** Submitted
With Maher Mamah, Youcef Mokrani, Bruno Sterner, and Nic Swanson.

2026 **Supersingular isogeny graphs and Hecke modules with level structure.** arXiv
With David Kohel.

2026 **Weber modular curves and modular isogenies.** arXiv
With David Kohel. To appear in the Polynesian Journal of Mathematics.

2025 **From orientations to ℓ-adic period vectors.** arXiv

2022 **On a modular approach to the OSIDH protocol.** Preprint
With David Kohel.

2020 **Orienting supersingular isogeny graphs.** arXiv
With David Kohel. Journal of Mathematical Cryptology, vol. 14, no. 1, pp. 414 - 437

SELECTED TALKS

complete list of talks available at
https://leonardocolo.com/talks.html

Jun. 2026 **Zero-Knowledge Proofs of Isogeny Diamonds.**
Ontario Cryptography Day (Hamilton, ON, CA).

Jan. 2026 **Orientations, modular symbols and p-adic periods for cryptography.**
JMM, Session on Mathematics of Isogeny-Based Cryptography (Washington, DC, USA).

Dec. 2025 **The modular symbol inversion problem.**
ATI Seminar – Arithmetic and Information Theory (Marseille, France).

Jul. 2024 **Supersingular isogeny graphs and Galois representations.**
William Tutte Colloquium (Waterloo, Canada).

Feb. 2023 **Oriented supersingular elliptic curves and class group actions.**
Algebraic and Combinatorial Methods for Coding and Cryptography (Marseille, France).

May 2021 **A modular approach to OSIDH.**
AGC^{2}T: Arithmetic, Geometry, Cryptography and Coding Theory (Marseille, France).

Jun. 2019 **Orienting supersingular isogeny graphs.**
Number-Theoretic Methods in Cryptology 2019 (Institut de Mathématiques de Jussieu, Paris, France).

TEACHING EXPERIENCE

complete list of courses available at
https://leonardocolo.com/teaching.html

2024–2026 **Course Instructor.** University of Waterloo.

Jan. 2025 **Group Instructor and Supervisor.** CIMPA School, Kampala, Uganda.

2019–2023 **Course Instructor.** Aix-Marseille Université.

2016–2017 **Course Instructor and Teaching Assistant.** Concordia University.

2016 **Teaching Internship.** Liceo Scientifico Leonardo da Vinci, Milan.
Internship supervised by Prof. Laura Chizzini.

WORKSHOPS

Aug. 2025 **Isogeny Graphs in Cryptography.** Banff International Research Station.
Group Leader for the project “Class-group action signatures”

SELECTED REFEREEING ACTIVITIES

2021 MathCrypt 2021 (Santa Barbara, USA).
2024 Algorithmic Number Theory Symposium XVI (Boston, USA).
2024 Eurocrypt 2025 (Madrid, Spain).
2025 Advances in Mathematics of Communications.

ACADEMIC ACTIVITIES

2025/2026 Reading group on Cryptography, University of Waterloo..
2024/2025 Crypto research group seminar, University of Waterloo..

PROFESSIONAL DEVELOPMENT

2026 **Information Security Specialist Program.** Fields Institute.
- Cryptography, network security, machine learning, quantum computing, risk management.

2026 **CryptoWorks21.** University of Waterloo.
- Cryptography, network security, quantum information, standards and certification, intellectual property protection, and commercialization.

Schools
- Arithmetic statistics (Marseille, France), May 2023.
- Cryptography and coding theory on curves and surfaces (Marseille, France), April 2019
- Mathematical foundations of asymmetric cryptography (Aussois, France), March 2019.
- ALGANT Summer school on Modular Forms (Padova, Italy), September 2017.

Conferences
- Quantum Consortium Day, February 2025 (University of Toronto, Canada).
- Algorithmic Number Theory Symposium, July 15–19, 2024 (Boston, USA)
- Aix-Marseille Cyber Security Forum, April 2021 & May 2022 (Marseille, FRA).

LANGUAGES

Mother tongue Italian.
Proficient English, French.

PROGRAMMING SKILLS

Familiar with HTML, CSS, JavaScript, Python, Sage, Magma, LaTeX, AWS, Terraform.
Certifications ECDL Certificate, HTML & CSS CCA Level 1.

OTHER SKILLS

Driving licence **Italy:** A2, B
Canada: G
Sport **Judo.** International competitions, Italian and Canadian national championships.